

SUPPLEMENT TO NEURAL BELLMAN OPERATORS

QIAN QI

Peking University

1. RELATION TO THE MAIN TEXT

This supplement expands the models and algorithms of the revised main manuscript. It does not introduce an alternative total loss or a different equilibrium definition. Policy evaluation freezes all actor outputs. Policy improvement freezes the critic's value and derivative channels. For games, only the acting player's objective is differentiated. For temporal selves, both exponential components evaluate the same future policy. The earlier main and supplementary sources are preserved in the revision archive for comparison; they are not alternative implementations of the revised method.

The inherited portfolio and quadratic laboratories retain their declared exact-structure classes. R6 trains generic multilayer policies in stopped consumption, preference adjustment, and dense nonlinear capital economies. R7 adds successful finite preference actors, neural game policies with full dynamic best responses, common-noise coverage records, and an independent arithmetic replay. R8 integrates those results and adds continuous-action training without maximizing labels, complete-action enclosures, direct and projection comparisons, and a separate continuous-transfer theorem. These approximation and verification classes are distinguished throughout the supplement.

2. POLICY EVALUATION AND THE RECURSIVE POLICY GRADIENT

Let a_ϕ be differentiable in a scalar parameter direction, and suppose V^{a_ϕ} is sufficiently differentiable for the following calculation. The terminal payoff is independent of the policy parameter. Assume differentiation under the expectation is justified, that the differentiated state and utility equations are integrable, and that the boundary conditions of the differentiated problem are zero. Denote the directional derivative of the policy value by w .

Differentiating the fixed-policy equation gives

$$0 = w_t + \mathcal{L}^{a_\phi} w + f_y(t, x, a_\phi, V^{a_\phi}) w + D_a \mathcal{H}^{a_\phi}[V^{a_\phi}] D_\phi a_\phi.$$

The operator derivative includes the dependence of both drift and covariance on the action. The derivative of the generator acting on the changing value is contained in $\mathcal{L}^{a_\phi} w$, while the action derivative of the generator acting on the fixed value is contained in $D_a \mathcal{H}$. No Hessian-action channel is discarded.

Applying the linear Feynman–Kac formula to the differentiated equation gives

$$w(t, x) = \mathbb{E}_{t,x}^{a_\phi} \int_t^\tau \exp \left(\int_t^s f_y(v, S_v, a_v, V^{a_\phi}) dv \right) D_a \mathcal{H}^{a_\phi}[V^{a_\phi}](s, S_s) D_\phi a_\phi(s, S_s) ds.$$

This formula concerns the performance derivative from the stated initial state. A domain-uniform Hamiltonian objective uses a different weighting. Such an objective is useful for approximate pointwise improvement, but it becomes a lifetime policy gradient only with the appropriate occupation and adjoint weights. Trajectories sampled by the current actor are detached in the implemented domain objective. Differentiating through their sampling law would define a different gradient estimator and must be analyzed separately.

The critic’s residual derivative is the derivative of the entire fixed-policy equation. In particular, its level is determined by the recursive evaluation equation together with boundary data. A separate negative-Hamiltonian gradient in the critic is not an innocuous level anchor: in the scalar stationary example it moves the minimizer away from the Bellman root.

3. REPRESENTATION, BOUNDARY CONDITIONS, AND ERROR MEASUREMENT

3.1. Smooth approximation and what it establishes

A derivative-approximation result concerns a smooth target on a compact domain and a family of networks whose capacity can increase. It does not imply that a fixed network attains zero residual, that optimization finds its best approximation, or that a nonsmooth target has classical derivatives everywhere. In a smooth problem, increasing the representation class can reduce approximation error in the value and derivatives. In a nonsmooth problem, NBO instead evaluates a consistent positive-

weight Bellman equation, or solves an explicitly regularized equation before taking a justified limit.

The hard-terminal architecture is $v(t, x) = G(x) + (T - t)N(t, x)$. It exactly enforces $v(T, x) = G(x)$ independently of training. A lateral Dirichlet face needs an additional compatible lifting and vanishing factor. For a reflected state, the Neumann condition is a different constraint. A loss on only interior points leaves all of these boundary questions unanswered. The regression test evaluates the terminal architecture at randomly chosen states and checks exact equality, not merely a small mean terminal error.

The recursive Epstein–Zin domain is $bV > 0$. An exponential output ensures positivity, while an interval transformation can keep bV between declared positive barriers. If $bG > 0$, one terminal-compatible positive representation is

$$bv(t, x) = \exp [\log(bG(x)) + (T - t)N(t, x)].$$

Its terminal value is exact. It does not by itself impose a lateral exit payoff or a two-sided barrier. These additional conditions must be built into the architecture or measured explicitly. Clipping an invalid utility argument inside a fractional power changes the equation and is not an acceptable hidden numerical repair.

3.2. From residual samples to a uniform bound

Let r be K -Lipschitz on a domain in $\mathbb{R}^d$. If every point is within distance h_V of a validation point, the triangle inequality gives $\|r\|_\infty \leq \max_{x \in \mathcal{V}} |r(x)| + Kh_V$. If K is estimated from samples rather than bounded, the resulting expression is itself a diagnostic, not a certified uniform bound.

An integral bound needs more assumptions. Suppose the sampling density is at least $p_0 > 0$, and each ball of radius r_0 or less centered in the domain intersects the domain in volume at least $c_D r^d$. Let $M = \|r\|_\infty$ and assume $M/(2K) \leq r_0$. Near a point approaching the maximum, Lipschitz continuity gives $|r| \geq M/2$ on a ball of radius $M/(2K)$. Hence

$$\mathbb{E}r^2 \geq \frac{p_0 c_D}{2^{d+2} K^d} M^{d+2}, \quad M \leq \left(\frac{2^{d+2} K^d \mathbb{E}r^2}{p_0 c_D} \right)^{1/(d+2)}.$$

For larger M , use the capped radius r_0 and the corresponding two-case inequality. A concentration argument is additionally needed to replace the population mean by an empirical one. This calculation explains why a dimension-free uniform certificate cannot be read directly from an average training loss.

The action gap is treated similarly. If an action net has covering radius h_A and the Hamiltonian is K_A -Lipschitz in the action, the unobserved supremum exceeds the net maximum by at most $K_A h_A$. Thus the net gap plus this term is an upper bound. A collection of successful deviations without a covering or duality argument is only a lower bound on exploitability.

4. BENCHMARK DERIVATIONS

4.1. Merton consumption and transversality

Write $b = 1 - \gamma$ and $V = KX^b/b$. Then $V_X = KX^{-\gamma}$ and $V_{XX} = -\gamma KX^{-\gamma-1}$. Substituting $c = mX$ into the stationary time-additive equation and dividing by the positive number KX^b gives

$$0 = \frac{m^b}{bK} - \frac{\rho}{b} + r + p(\mu - r) - m - \frac{1}{2}\gamma p^2 \sigma^2.$$

Consumption maximization gives $m^{-\gamma} = K$; portfolio maximization gives $p = (\mu - r)/(\gamma\sigma^2)$ when interior. With a no-short-sale constraint and negative excess return, the maximizing share is zero. Let $R_* = r + p(\mu - r) - \gamma p^2 \sigma^2/2$. After substitution, $\gamma m = \rho + (\gamma - 1)R_*$.

For the optimal constant shares, $e^{-\rho t} \mathbb{E}[X_t^{1-\gamma}]$ decays as e^{-mt} . Thus $m > 0$ gives the candidate's additive transversality condition. Verification against a larger policy class also requires the relevant localization and integrability conditions for the competitors, as specified in the main text. Policies with minus-infinite utility are not preferred to the finite candidate. The numerical laboratory checks the candidate formulas and does not silently apply a stationary constant to an unspecified finite-horizon terminal problem.

4.2. Recursive coefficient and the utility interval

For $f(c, V) = \rho[c^a(bV)^{1-a/b} - bV]/a$, the same homothetic substitution gives

$$0 = \frac{\rho}{a}[m^a K^{-a/b} - 1] + R_* - m.$$

Since $f_c = V_X$ implies $\rho m^{a-1} K^{-a/b} = 1$, the first term becomes $(m - \rho)/a$. Consequently, $m = \psi\rho + (1 - \psi)R_*$ and $K = [\rho m^{a-1}]^{b/a}$. The formula assumes the stationary candidate has positive consumption. It is not an infinite-horizon utility-selection theorem.

For the finite-horizon matching-bequest test at $a > 0 > b$, define $Y = bV$. The consumption domain is compact and bounded away from zero. At sufficiently small positive Y , $c^a Y^{-a/b} < 1$ uniformly in consumption, so $f < 0$. At sufficiently large Y , the reverse inequality holds and $f > 0$. Because dividing by negative b reverses the order of the utility endpoints, the large- Y constant is a lower utility barrier and the small- Y constant is an upper utility barrier. The matching bequest lies between them after enlarging the interval. Lipschitz scalar comparison then keeps each policy utility inside the interval, on which the aggregator and its utility derivative are bounded. This verifies the finite-horizon interpretation without claiming an unrestricted infinite-horizon result at those parameters.

5. ENDOGENOUS-PREFERENCE DETAILS

5.1. State restrictions and the reference experiment

In the baseline application, the risk-aversion interval lies strictly above one and the process is reflected. The reference wealth problem is stopped at positive finite bounds. Consumption shares are bounded away from zero, adjustment efforts are bounded, and the horizon is finite. These choices bound the flow utility, its relevant derivatives away from the terminal-boundary compatibility issue, and the bequest. They provide a well-defined finite reference calculation. The numerical reflection is a folded reflection of Brownian cubature endpoints; the wealth boundary is absorbing with the prescribed payoff. The same terminal and boundary values are imposed for both adjustment-cost parameters.

The grid reference operates in $y = \log X$. Its drift is

$$b_y = r + p(\mu - r) - m - \frac{1}{2}p^2\sigma_X^2,$$

and its covariance with u is $\varrho\sigma_u p\sigma_X$. The two Brownian-factor columns can be written $(\sigma_u, \varrho p\sigma_X)^\top$ and $(0, \sqrt{1 - \varrho^2}p\sigma_X)^\top$. Four symmetric cubature directions reproduce this covariance. The correction $-p^2\sigma_X^2/2$ in log wealth is essential; omitting it would change the economic diffusion.

For each action, the backward update is current utility times the time step plus the exponentially discounted positive-weight interpolation of next-period value. Feasible actions are enumerated on the declared 27-element grid. Because the terminal/exit payoff is fixed, raising k weakly lowers each action's payoff and, by backward induction, lowers value at every state and time. The recorded arrays permit a direct check of this order. The reported grid-refinement difference is not an estimate of unrestricted neural approximation error.

5.2. Policy conditions and comparative statics

The consumption marginal utility in reference units is $U_c = \bar{c}^{-1}(c/\bar{c})^{-u}$. Equating it to V_X yields $c = \bar{c}(\bar{c}V_X)^{-1/u}$. Adjustment maximizes $\theta V_u - k\theta^2/2$ and hence uses the projection of V_u/k under a box. Portfolio choice maximizes a quadratic with coefficient $X^2\sigma_X^2 V_{XX}/2$. The projected first-order solution is valid only when this coefficient is negative. These conditions clarify why numerical curvature and feasibility must be checked independently.

The cross derivative V_{uX} is an endogenous object. Its sign cannot be inferred from the label "risk aversion" or from a negative shock correlation. Nor can the sign of $\partial_k \theta^k$ be inferred by differentiating $1/k$ while keeping V_u^k fixed. In contrast, the value monotonicity in k holds policy by policy. At a differentiable optimum the envelope formula follows because k appears only in the quadratic running cost. Changing the normalization function $\kappa(u)$ can change both adjustment and portfolio policies even while leaving utility curvature in consumption unchanged.

An observation model is needed for structural estimation. With latent u , uncertainty in κ , and unknown adjustment or shock parameters, a sensitivity plot for

one policy component does not prove identification. A practical estimation exercise would compute moment derivatives jointly, inspect their rank after normalization restrictions, and propagate numerical approximation error into the implied moment map. Those are distinct tasks from solving the controlled economy.

6. TEMPORAL-SELF DEVIATIONS

For the two-exponential kernel, the component value under constant consumption share m is $W_j(X) = m^b X^b / [b\lambda_j(m)]$, where $\lambda_j(m) = \rho_j - b(r - m)$. A deviation to share z for duration ϵ , followed by m , has component value at $X = 1$

$$J_j(z, \epsilon) = \frac{1}{b} \left[z^b \frac{1 - e^{-\lambda_j(z)\epsilon}}{\lambda_j(z)} + e^{-\lambda_j(z)\epsilon} \frac{m^b}{\lambda_j(m)} \right].$$

The continuous extension is used if $\lambda_j(z) = 0$. The deviation gain is $\sum_j w_j J_j(z, \epsilon) - \sum_j w_j W_j(1)$, with weights $(\beta, 1 - \beta)$. This is the quantity independently maximized in the experiment, rather than a restatement of the equilibrium first-order condition.

At an equilibrium share, the first-order gain is nonpositive for every feasible instantaneous share and equals zero at the equilibrium share. A finite-duration self can generally choose a slightly different share, so the maximized gain need not be exactly zero for positive duration. The required diagnostic is its gain divided by duration as duration tends to zero. For $\beta = 1$ the ordinary exponential solution is recovered.

As κ grows, the kernel becomes increasingly concentrated near the present before approaching its long-run component. Taking the instantaneous-deviation limit and taking $\kappa \rightarrow \infty$ are conceptually different operations. The finite- κ model in this paper explicitly specifies the order: the spike condition is derived for fixed κ , and any duration-limit comparative static is taken afterward. This avoids assigning an undefined continuous-time meaning to a discrete present-bias multiplier.

7. GAMES AND INDEPENDENT BEST RESPONSES

A fixed rival profile turns each player's problem into an ordinary controlled Markov problem, so the main verification results apply player by player. The derivative graph must implement this reduction. During player i 's actor step, every

rival policy and every derivative of player i 's critic is detached. During player i 's critic step, all players' actions are detached. A sum of these reporting losses can be logged, but differentiation of a joint-profit objective is not equivalent to those block steps.

The two-firm static diagnostic has best response $q_i = (1 - q_j)/2$ when interior. At the joint-profit allocation $(1/4, 1/4)$, the deviating firm chooses $3/8$ and improves profit from $1/8$ to $9/64$, a gain of $1/64$. The own-action derivative is $1/4$ at the joint optimum. These numbers verify a graph-level distinction; they are not a computation of the dynamic capital game.

For the dynamic problem, a reported equilibrium should include an independent solution of each fixed-rival best-response problem, or an upper bound on the unilateral Hamiltonian gap over the entire verification domain. A finite list of unsuccessful deviations does not supply such an upper bound. Multiple initializations, asymmetric profiles, boundary actions, and the market-size normalization are part of the same experiment. The specification in the main text retains the dynamic Cournot application but does not invent a dynamic Nash table from the static arithmetic.

8. STOCHASTIC HESSIAN OBJECTIVES AND COMPLEXITY

Let $B = \Sigma^\top D^2 v \Sigma$, a symmetric matrix in Brownian-factor space. For Gaussian probes, Isserlis' identity gives $\mathbb{E}(z^\top B z) = \text{tr } B$ and $\text{Var}(z^\top B z) = 2 \text{tr}(B^2) = 2 \|B\|_F^2$. Averaging K independent probes divides the variance by K . Rademacher probes instead have variance $2 \sum_{i \neq j} B_{ij}^2 / K$. The variance depends on the chosen covariance factor and probe law, both of which must be specified.

Write $\hat{q}_j = q + \epsilon_j$, with conditionally mean-zero independent errors. Then

$$\mathbb{E}[(d - \hat{q}_1/2)(d - \hat{q}_2/2)] = (d - q/2)^2 + \frac{1}{4} \mathbb{E}[\epsilon_1 \epsilon_2] = (d - q/2)^2.$$

Using one bank twice would retain its variance term. In the scalar Gaussian example, a single-bank squared residual has expectation $1 - h + 3h^2/4$, minimized at $h = 2/3$, whereas the exact residual $(1 - h/2)^2$ is minimized at $h = 2$. The independent product restores the latter population objective. The saved Monte Carlo diagnostic

uses 400,000 pairs and records both empirical estimates; it does not claim that the sampling error is identically zero.

The core implementation uses exact Hessians to make the covariance contraction independently testable. A Hessian-vector implementation can replace it without changing the conditional-independence requirement. The cost of a Hessian-vector product depends on the network graph, not just the state dimension. Total cost also includes sampling, optimization, domain coverage, action maximization, and stopping at an agreed error level. A quadratic-neural representation with a Lyapunov solve has different approximation and optimization costs from a generic multi-layer network. The inherited quadratic laboratory is explicitly the former; the new dense nonlinear experiments actually train generic networks and Hessian-vector estimators.

9. REPRODUCTION AND FURTHER COMPARISONS

From the repository root, execute the experiment program and then the regression suite. The dependency versions and CPU setting are recorded with the results. The experiment output distinguishes acceptance of a calibration or identity test from execution of a reference computation. A separate reviewer-check output reproduces the original twelve diagnostics against the archived original manuscript, not against the revised article.

A broader method comparison should use one problem specification and evaluate value error, policy error, boundary error, and the independent action gap at matched accuracy. A direct neural-HJB comparator eliminates the actor with the same feasible maximization routine. A temporal-regression comparator uses the same recursive discretization and utility domain, without adding discount twice. A separability-aware classical comparator exploits all available economic structure. Network capacity, sampling, covariance treatment, random seeds, failures, wall time, and memory should all be reported. R6 executes the architecture-matched direct-HJB and finite-Bellman comparisons. The broader deep-BSDE comparison remains unexecuted and is not counted as a measured advantage. R7 neural-game evidence and the additional R8 comparisons are reported in the integrated sections below.

The inherited read-only workflow is preserved. The separately scoped R6 replication workflow commits materialized source and raw recomputation outputs only to its designated candidate branch, without editing the review or earlier revision branches. Hashes and exact source commits distinguish these artifacts. No workflow rewrites a failed numerical accuracy target as a pass.

10. IMPLEMENTED ERROR ACCOUNTS AND ARITHMETIC

The implemented map returns a candidate and a verification record, rather than assigning authority to an optimizer’s exit code. In the differential consumption economy, the record identifies the binary64 weights, both stopping faces, all dyadic leaves, global consumption maximization, and the error bounds for the continuous diffusion. In NDU it instead identifies the finite state–action–time model, its positive transition weights, and time-indexed evaluation and improvement errors. The dynamic-game record fixes the entire rival policy before an independent backward best-response solve. An unchecked continuous-action or time-discretization term is represented as unavailable in the machine-readable output, not as zero.

For finite time levels, let $\delta_t = \|v_t - T_t^a v_{t+1}\|_\infty$ and $\epsilon_t = \|T_t v_{t+1} - T_t^a v_{t+1}\|_\infty$. If both maps are β_t -Lipschitz, the triangle inequality gives

$$\|v_t - V_t^a\| \leq \delta_t + \beta_t \|v_{t+1} - V_{t+1}^a\|, \quad \|v_t - V_t^*\| \leq \delta_t + \epsilon_t + \beta_t \|v_{t+1} - V_{t+1}^*\|.$$

Summing the two inequalities yields the regret account. The same recurrence with a player’s unilateral gap bounds the gain from a dynamic deviation when that player’s policy evaluation is exact. Accounts must be checked at each time level: a time-zero bound need not dominate a later-time bound when discounting and errors differ. The coarse-game regression exercises this distinction and verifies that a missing pure equilibrium is reported rather than silently treated as exact.

10.1. Full-domain interval implementation

An interval $[l, u]$ encloses an exact real quantity. Each elementary operation expands its rounded endpoints with the adjacent binary64 number. Affine layers split positive and negative weights and include a conservative dot-product forward-error allowance. The arithmetic contract requires binary64 round-to-nearest arithmetic,

the stated dot-product error bound, and absence of exceptional overflow or flush-to-zero behavior. Stored parameters denote their exact binary64 real values; the verifier does not assert an unrecorded decimal model identity.

For an exponential argument x , range reduction gives $r = x/2^s$ with $|r| \leq 1/8$. A degree-fourteen Taylor polynomial has remainder bounded by

$$e^{|r|} |r|^{15}/15! < 2(1/8)^{15}/15! < 2^{-80}.$$

Interval Horner evaluation and repeated squaring enclose e^x . The identity $\tanh x = 1 - 2/(e^{2x} + 1)$ then supplies $\tanh$ values and derivative channels $1 - \tanh^2 x$ and $-2 \tanh x(1 - \tanh^2 x)$. Powers of two and the range-reduction domain are checked in the code.

For logarithms, exact binary decomposition gives $x = 2^k m$, with $1 \leq m < 2$. Writing $z = (m - 1)/(m + 1)$,

$$\log m = 2 \sum_{j=0}^{24} \frac{z^{2j+1}}{2j+1} + R, \quad |R| \leq \frac{2|z|^{51}}{51(1-z^2)} < 2^{-72}$$

for the outward-enclosed range $|z| < .334$. The same expansion encloses $\log 2$. Neither construction presumes correctly rounded library $\tanh$ or logarithm functions. The saved test suite checks transcendental endpoints against eighty-digit arithmetic, differentiates the trained network independently, and checks all certificate hashes and leaf adjacencies. These tests supplement, but do not replace, the arithmetic contract.

The scalar action objective $h(c) = \log c - pc$ has second derivative $-1/c^2$. If C encloses both the actor and its maximizing alternative, strong concavity on their joining segment gives

$$h(c^*) - h(c) \leq h'(c)(c^* - c) - (c^* - c)^2/(2C^2) \leq C^2 h'(c)^2/2.$$

The argument includes constrained endpoints. Taking the minimum with the natural interval action-gap enclosure markedly reduces verification cost, without using a sampled gradient as a bound. Residual targets are decided on every leaf; reaching a maximum subdivision depth does not change a failed target into a pass.

10.2. *Application-specific hypotheses*

The stopped consumption diffusion is uniformly nondegenerate on its compact domain; controls and coefficients are bounded and its endpoint data are continuous. The smooth networks and strictly positive consumption map meet the regularity and log-domain requirements of the verification argument. Both stopping-face errors and the full action interval are checked. The upwind reference error is not a premise of its continuous certificate.

The NDU computation is a positive-weight, finite-action, finite-horizon economy with prescribed discrete stopping and folded preference transitions. Positivity, all terminal and wealth-face values, and every finite action are evaluated. The continuous reflected PDE's corner compatibility, continuous-action coverage error, and diffusion approximation error are not inferred from these tests. Expanding a feasible box changes the constrained economy; densifying a fixed box does not. The sensitivity record keeps these operations separate.

The dense capital computation uses bounded nonlinear drift in log coordinates, bounded positive consumption fractions, constant diffusion, and finite horizon. Its flow has at most linear log-state growth and its terminal payoff at most quadratic growth; these are integrable under the declared bounded-drift diffusion. The fitted box is a sampling domain, not a state constraint. Its sampled differential residuals do not check comparison barriers or uniform errors on $\mathbb{R}^d$. Euler payoff estimates carry both sampling error and unbounded-state/time-discretization issues not covered by the reported Monte Carlo standard error.

The dynamic capital game is a fully enumerated finite approximation with stopped joint boundaries. Stage equilibria and dynamic fixed-rival best responses are numerical statements about that model. The continuous-control best-response comparison assumptions in the main theorem remain logically distinct. The retained homothetic recursive-utility and temporal-self examples use their stated utility domains, bequests, and integrability conditions; R6 does not relabel them as unrestricted neural experiments.

11. DESIGN, BUDGETS, AND REPRODUCIBILITY

The stopped-consumption study uses float64 and the existing generic derivative core. Identical seed labels fix initialization and collocation construction across methods. L-BFGS iteration caps are matched, but line-search closure counts need not match. Those counts, fit times, and separate verification times are saved. The grid reference uses exact maximization within each upwind drift region, not the learned actor. Reference comparisons omit four endpoint-adjacent nodes only for finite-difference derivative comparisons; all endpoints enter the boundary check and the interval certificate.

The NDU actor predicts one of the finite feasible triples. Its normalized Bellman-loss objective gives economically costly action mistakes more weight than ordinary classification would. Absorbing boundary rows are excluded only from actor fitting, because their actions have no effect; they remain in policy evaluation and verification. The failed initial classification implementation is archived with its actual outputs. Principal seed 11 misses the declared policy-regret target even after the improvement; its result is not discarded. The direct finite-Bellman comparator uses the same critic class and total inner-iteration cap.

The dense-capital primary study uses exact Hessians; the trace follow-up actually backpropagates through Hessian-vector products. Product and single-bank implementations use four probes in total, but only the product splits them into conditionally independent banks. The Monte Carlo regression separately checks the value and gradient of the product objective with tolerances .012 and .025. An unbiased objective can have higher variance, and no claim of superior finite-sample accuracy follows merely from its expectation. The covariance in the retained experiment is modest; the reported results reflect that scale.

The initial high-dimensional state box was too narrow to certify the realized policy paths. The wider-box design is therefore identified as a follow-up, not a preregistered experiment. Forty- and eighty-step policy simulations share the same seed convention across methods at a given resolution. They are not a coupled Brownian extrapolation across resolutions. Raw per-path payoffs allow paired

policy comparisons, but neither a Monte Carlo confidence interval nor a critic-
payoff agreement proves closeness to the optimal value.

Each executed program records all declared seeds, its budget, residual or deviation arrays, and an explicit scope. Process peak resident memory is a high-water mark of the process, potentially including earlier runs. It is not an isolated memory-complexity measurement for an individual method. CPU timings are execution-specific and do not reproduce the historical accelerator claims. The independent raw-output workflow binds its source commit and file hashes; the source archive keeps the R2 manuscript and supplement byte-for-byte for comparison.

12. RUN-LEVEL NUMERICAL ACCOUNTS

12.1. *Consumption fitting and verification*

Method	Seed	Width/depth	Fit seconds	Verify seconds	Pass
nbo	11	16/2	2.1790	1.0822	True
nbo	29	16/2	2.0422	1.4045	True
nbo	47	16/2	2.0550	1.3100	True
direct	11	16/2	1.9767	0.4974	True
direct	29	16/2	0.8018	0.4875	True
direct	47	16/2	2.1912	0.5481	True
joint	11	16/2	0.2578	–	False
joint	29	16/2	2.5836	–	False
joint	47	16/2	2.5935	–	False
nbo	11	8/2	2.5767	–	True
nbo	11	32/2	1.6599	–	True
nbo	11	16/1	1.6198	–	True
nbo	11	16/3	2.3456	–	True
nbo	11	16/2	2.1977	–	True
nbo	11	16/2	1.8599	–	True

Pass in this table means the prespecified sampled diagnostic. Continuous certification is reported separately and only for the primary NBO and direct-HJB runs. Every closure count, pointwise derivative array, parameter, and iteration trace is in the corresponding raw result.

12.2. Action bounds and numerical domains

Actions	Grid/time	k	Center value	Upper m freq.	Upper p freq.
r2	17x25/20	1.0000	-11.9152	0.9994	0.7885
r2	17x25/20	5.0000	-11.9570	0.9992	0.7890
adjustment	17x25/20	1.0000	-10.6081	1.0000	0.8364
adjustment	17x25/20	5.0000	-10.9202	1.0000	0.8405
expanded	17x25/20	1.0000	-4.8410	0.9898	0.5440
expanded	17x25/20	5.0000	-4.9243	0.9855	0.5515
refined	17x25/20	1.0000	-4.8406	0.9737	0.5301
refined	17x25/20	5.0000	-4.9214	0.9688	0.5353
expanded_domain	17x25/20	1.0000	-2.7504	0.9698	0.5665
expanded_domain	17x25/20	5.0000	-2.8452	0.9499	0.6068
expanded	17x25/20*	1.0000	-5.0185	0.9997	0.3164
expanded	17x25/20*	5.0000	-5.1095	0.9971	0.3251
expanded	25x37/40	1.0000	-4.8395	0.9902	0.5738
expanded	25x37/40	5.0000	-4.9228	0.9853	0.5787
expanded	33x49/80	1.0000	-4.8302	0.9872	0.6050
expanded	33x49/80	5.0000	-4.9143	0.9823	0.6102

An asterisk marks wealth interval $[\cdot 25, 4]$, rather than $[\cdot 5, 2]$. Frequencies use all nonabsorbing state–time nodes. The raw record includes both bounds for each of the three controls. “Expanded has 175 actions; “refined has 1,053 actions in the same box; “expanded_domain changes the box itself.

12.3. Dense neural runs, trace estimators, and coverage

Method	d/seed	Steps	RMS	Gap	Seconds	Pass
exact	2/11	600	0.0125	1.44×10^{-4}	2.7329	True
direct	2/11	600	0.0123	0	3.4870	True
exact	2/29	600	0.0161	2.28×10^{-4}	2.6554	True
direct	2/29	600	0.0161	0	3.4759	True
exact	2/47	600	0.0132	2.42×10^{-4}	2.6361	True
direct	2/47	600	0.0136	0	3.5572	True

	Method	d/seed	Steps	RMS	Gap	Seconds	Pass	
1								1
2	exact	5/11	600	0.0101	0.0012	4.1518	True	2
3	direct	5/11	600	0.0101	0	5.0391	True	3
4	exact	5/29	600	0.0131	8.38×10^{-4}	4.2150	True	4
5	direct	5/29	600	0.0129	0	4.9992	True	5
6	exact	5/47	600	0.0122	0.0020	4.1258	True	6
7	direct	5/47	600	0.0118	0	5.0485	True	7
8	exact	10/11	600	0.0163	0.0052	6.8652	True	8
9	direct	10/11	600	0.0138	0	7.6139	True	9
10	exact	10/29	600	0.0188	0.0035	6.9393	True	10
11	direct	10/29	600	0.0168	0	7.7251	True	11
12	exact	10/47	600	0.0141	0.0050	6.8241	True	12
13	direct	10/47	600	0.0139	0	7.6988	True	13
14	exact	20/11	600	0.0148	0.0113	12.8180	False	14
15	direct	20/11	600	0.0135	0	13.5001	True	15
16	exact	20/29	600	0.0138	0.0090	12.6025	True	16
17	direct	20/29	600	0.0129	0	13.4929	True	17
18	exact	20/47	600	0.0320	0.0114	12.9103	False	18
19	direct	20/47	600	0.0170	0	13.6205	True	19
20	product	10/11	600	0.0163	0.0052	3.9801	True	20
21	single	10/11	600	0.0162	0.0052	3.9075	True	21
22	product	10/29	600	0.0187	0.0035	3.9346	True	22
23	single	10/29	600	0.0186	0.0034	3.9263	True	23
24	product	10/47	600	0.0141	0.0050	3.9233	True	24
25	single	10/47	600	0.0140	0.0050	3.9264	True	25
26	exact*	10/11	1000	0.0236	0.0019	11.1414	True	26
27	direct*	10/11	1000	0.0278	0	12.4295	False	27
28	exact*	10/29	1000	0.0249	0.0022	10.9333	True	28
29	direct*	10/29	1000	0.0276	0	12.3312	False	29
30	exact*	10/47	1000	0.0281	0.0026	11.0150	False	30
31	direct*	10/47	1000	0.0293	0	12.3768	False	31
32	exact*	20/11	1000	0.0200	0.0040	20.8834	True	32
	direct*	20/11	1000	0.0211	0	22.9271	True	
	exact*	20/29	1000	0.0200	0.0034	20.0811	True	
	direct*	20/29	1000	0.0227	0	21.9897	True	
	exact*	20/47	1000	0.0216	0.0071	19.9588	True	

Method	d/seed	Steps	RMS	Gap	Seconds	Pass
direct*	20/47	1000	0.0244	0	21.7639	True

An asterisk marks the follow-up fitting domain $[-1.5, .5]^d$ and 1,000 updates. Product uses two independent two-probe banks; single squares a four-probe trace. Every row retains its own history and exact-trace diagnostic set. Runtime is not a hardware-independent complexity bound.

12.4. Policy-payoff diagnostics

Method	d/seed	Critic at zero	Euler mean	MC s.e.	Critic minus mean
exact	10/11	-1.1925	-1.3045	0.0041	0.1120
direct	10/11	-1.2054	-1.2729	0.0036	0.0675
exact	10/29	-1.1781	-1.2999	0.0040	0.1218
direct	10/29	-1.1899	-1.2701	0.0036	0.0802
exact	10/47	-1.2080	-1.2967	0.0040	0.0887
direct	10/47	-1.2077	-1.2698	0.0036	0.0621
exact	20/11	-1.1809	-1.3458	0.0041	0.1649
direct	20/11	-1.1986	-1.2834	0.0035	0.0848
exact	20/29	-1.1920	-1.2895	0.0035	0.0975
direct	20/29	-1.2071	-1.2754	0.0034	0.0684
exact	20/47	-1.0343	-1.5052	0.0036	0.4709
direct	20/47	-1.1661	-1.3051	0.0036	0.1390
exact*	10/11	-1.2974	-1.2521	0.0035	-0.0453
direct*	10/11	-1.2992	-1.2527	0.0035	-0.0465
exact*	10/29	-1.2933	-1.2521	0.0035	-0.0412
direct*	10/29	-1.2871	-1.2521	0.0035	-0.0350
exact*	10/47	-1.2939	-1.2527	0.0035	-0.0412
direct*	10/47	-1.2959	-1.2527	0.0035	-0.0433
exact*	20/11	-1.2788	-1.2646	0.0034	-0.0142
direct*	20/11	-1.2787	-1.2656	0.0034	-0.0132
exact*	20/29	-1.2712	-1.2646	0.0034	-0.0066
direct*	20/29	-1.2563	-1.2689	0.0034	0.0126
exact*	20/47	-1.2772	-1.2645	0.0034	-0.0127

Method	d/seed	Critic at zero	Euler mean	MC s.e.	Critic minus mean
direct*	20/47	-1.2821	-1.2652	0.0034	-0.0169

These estimates use 2,048 paths and 80 Euler steps from zero log capital. Forty-step samples are also saved. Identical random seeds pair policy comparisons at the same step size; the forty- and eighty-step simulations are not a Brownian-coupled refinement estimator. The standard error measures sampling error only. The initial critic is not an external optimal-value reference.

13. PROOF AND IMPLEMENTATION OF THE CONTINUOUS-ACTION ENVELOPE

We use the notation of the main text's budget-safe envelope proposition. The nodal economy is fixed before verification. Its terminal and stopping payoffs, primitive coefficients, interpolation abscissae, and compact action box are part of the object being bounded. This is stronger than checking many action samples and weaker than verifying the continuous-state diffusion without a discretization account.

13.1. Complete-cover invariant and backward proof

At a given state and time, the initial action box covers the entire feasible set. Bisection replaces one closed box by two closed boxes with the same union. Their shared face is harmless. Induction on the number of subdivisions preserves coverage. A pruned box is retained logically through the inequality $u_B \leq c_i + \tau$, where u_B is an outward upper enclosure on that whole box, and c_i is a lower enclosure at a feasible incumbent. If the incumbent later increases, the previous pruning inequality remains true. Consequently every feasible action is either in an unresolved box or in a pruned box with payoff at most the current threshold. Taking the maximum of the threshold and all unresolved upper bounds dominates the action supremum, whether or not the supremum is attained. Termination on a numerical budget does not change this argument.

At N , the terminal inequalities are given. If $V_{n+1}^* \leq U_{n+1}$, positivity gives $T_n V_{n+1}^* \leq T_n U_{n+1} \leq U_n$. If $L_{n+1} \leq V_{n+1}^\pi$, the same argument gives $L_n \leq T_n^\pi L_{n+1} \leq$

$T_n^{\pi_n} V_{n+1}^{\pi} = V_n^{\pi}$. Feasibility implies $V_n^{\pi} \leq V_n^*$. This proves the main envelope inequality. Substochasticity further gives

$$\|T_n w - T_n z\|_{\infty} \leq q_n \|w - z\|_{\infty}.$$

Applying this inequality to $U_n - T_n V_{n+1}^*$ yields the stated excess recursion and its product-sum expansion. The proof uses $q_n > 0$, not a stationary contraction assumption. Outward addition is used for the pruning threshold and final subtraction. The implementation records a requested tolerance and a retained optimization bracket separately. A budget-exhausted bracket is never replaced by that tolerance. *Q.E.D.*

13.2. Interpolation, reflection, stopping, and arithmetic

For a cell with coordinates $(u_0, u_1) \times (y_0, y_1)$, the continuation is

$$Iw(u, y) = (1 - s)(1 - t)w_{00} + s(1 - t)w_{10} + (1 - s)tw_{01} + stw_{11},$$

$$s = \frac{u - u_0}{u_1 - u_0}, \quad t = \frac{y - y_0}{y_1 - y_0}.$$

For fixed u , this is affine in y , so its extremes on a subrectangle occur at the two y endpoints. At either endpoint it is affine in u , so the extremes occur at the two u endpoints. Applying this argument to every intersected cell proves the corner enclosure. Nonnegative interpolation weights also permit interval nodal inputs: evaluating the corner expressions outward encloses every admissible nodal realization. A separate valid enclosure evaluates the rectangle center and adds cell-slope bounds over every intersected cell. Slopes are computed from stored nodal differences, not estimated from random derivative samples.

Preference reflection is piecewise affine. Its extrema on an interval occur at the interval endpoints or a crossed folding point. The allowed one-step movement is checked to exclude unhandled multiple folds. For log wealth, boxes fully inside the domain use interpolation, boxes fully outside use the prescribed clipped exit payoff, and boxes crossing an exit face use the union of both. On the stored grid the boundary nodes equal the exact prescribed payoff; at an off-grid exit the analytic payoff need not equal its nodal interpolant. The union construction avoids an unjustified global smoothness or Lipschitz assumption across that switch.

The implementation inherits the archived outward elementary and transcendental routines. Products, quotients, powers through logarithm and exponential, and endpoint rounding are enclosed. The logarithm's input is bounded below by .02; the preference denominator is separated from zero because $u \in [1.2, 3]$. The recorded control model treats the stored binary64 primitive coefficients and abscissae as exact real inputs. The evidence also tests agreement with the inherited training evaluator on the finite action grid. Such a regression check is useful for implementation consistency but is not the proof of global action coverage. That proof is the maintained union invariant and the interval inclusion property. The arithmetic account is conditional on the specified floating-point execution model, not a formal verification of hardware, libraries, or the Python interpreter.

13.3. *A common upper envelope for multiple policies*

The optimal upper array depends on the economy, not on which feasible incumbent produced the lower search values. Thus the completed upper array can be reused for any frozen feasible policy on the same state, time, and continuous-action domains. We separately propagate each policy's lower and upper payoff arrays and subtract its lower array from the common optimal upper array. Costs are reported both as shared verification and as the cost of an individual policy evaluation. Sharing a verifier does not make its cost vanish. It also does not justify transferring the upper array to a changed wealth box, action constraint, utility normalization, time step, or interpolation rule.

14. SEPARATING NUMERICAL AND CONTINUOUS-ECONOMY ERROR

The following result specifies the transfer work needed in addition to a nodal certificate. It is not used to set an uncomputed transfer term to zero. Let $\mathcal{T}_n$ be the exact economic one-step optimality operator with its continuously monitored stopping and reflection rule, and let $\mathcal{T}_n^\pi$ be the operator for a declared continuous-state implementation of the numerical policy. Let I_n be a positive, sup-norm-nonexpansive interpolation from grid values to the economic state domain. Suppose the exact operators are q_n -Lipschitz on the comparison class.

PROPOSITION 1—Transfer with separate optimal and policy defects: *Suppose terminal interpolation error is at most B and, along the relevant nodal continuations,*

$$\|\mathcal{T}_n I_{n+1} v_{n+1}^* - I_n T_n v_{n+1}^*\|_\infty \leq d_n^*,$$

$$\|\mathcal{T}_n^\pi I_{n+1} v_{n+1}^\pi - I_n T_n^\pi v_{n+1}^\pi\|_\infty \leq d_n^\pi.$$

Define $D_N^* = D_N^\pi = B$ and $D_n^j = d_n^j + q_n D_{n+1}^j$ for $j \in \{*, \pi\}$. Then the exact economic values satisfy

$$0 \leq \mathcal{V}_n^* - \mathcal{V}_n^\pi \leq I_n(U_n - L_n) + D_n^* + D_n^\pi.$$

The action-domain approximation belongs in d_n^* , while policy implementation, boundary monitoring, state interpolation, and time/cubature errors may affect both defects.

PROOF: Insert $\mathcal{T}_n I_{n+1} v_{n+1}^*$ between the exact and interpolated nodal Bellman values. The Lipschitz property bounds the propagated value difference by $q_n D_{n+1}^*$, and the displayed defect bounds the remaining difference by d_n^* . Backward induction gives $\|\mathcal{V}_n^* - I_n v_n^*\| \leq D_n^*$. The same argument gives the fixed-policy bound. Subtract the two decompositions and use positivity of I_n and $v_n^* - v_n^\pi \leq U_n - L_n$. Feasibility gives the lower inequality. *Q.E.D.*

The proposition can use local regularity, coupling, barrier, or weak-approximation estimates to bound its defects; it does not claim that those hypotheses hold automatically for a fitted critic near a stopped boundary. In the present two-state computations, action coverage is operationally bounded on the nodal economy. The continuous-state/time defects remain explicitly uncomputed. Nested grids and common frozen policies diagnose sensitivity but do not provide the one-sided inequalities required by this proposition.

TABLE V: Frozen continuous-policy refinement, common economic domain

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	11	17×25	20	-4.843442	-4.841008
Actor	11	25×37	40	-4.848086	-4.839462

Method	Seed	Grid	Steps	Center payoff	Center reference
Actor	11	33×49	80	-4.842578	-4.830165
Actor	29	17×25	20	-4.841652	-4.841008
Actor	29	25×37	40	-4.846754	-4.839462
Actor	29	33×49	80	-4.840763	-4.830165
Actor	47	17×25	20	-4.844135	-4.841008
Actor	47	25×37	40	-4.848828	-4.839462
Actor	47	33×49	80	-4.843387	-4.830165
Direct	11	17×25	20	-4.843773	-4.841008
Direct	11	25×37	40	-4.857445	-4.839462
Direct	11	33×49	80	-4.852920	-4.830165
Direct	29	17×25	20	-4.841889	-4.841008
Direct	29	25×37	40	-4.854488	-4.839462
Direct	29	33×49	80	-4.848755	-4.830165
Direct	47	17×25	20	-4.844185	-4.841008
Direct	47	25×37	40	-4.854178	-4.839462
Direct	47	33×49	80	-4.848680	-4.830165

Notes. Policies are frozen on the original time schedule and bilinearly interpolated in state. References still use 175 actions. Differences are sensitivity diagnostics, not continuous-state/time upper bounds.

15. STRATEGIC DEVIATIONS AND MISSING PURE STAGES

For a fixed Markov rival profile in a finite-horizon finite-state/action game, condition on any history. The own player's conditional payoff from a randomized current action is a convex combination of pure-action payoffs with the same continuation best-response values. It cannot exceed the largest such payoff. Backward induction therefore bounds every randomized and history-dependent unilateral policy by the Markov pure-action best-response recursion. Outward propagation then provides a bound over this full deviation class, not merely the particular local deviations observed during actor training.

Existence of a pure equilibrium in a fitted one-step game is a different matter. If none exists, minimum-defect selection produces a feasible profile, not an exact equilibrium. The independent dynamic best response still evaluates that profile correctly. The following records give every missing pure stage in the six principal R7 neural runs, including time index, state, and minimum maximum stage defect. Indices begin at zero. The six runs have $6 \times 30 \times 225 = 40,500$ active state-time nodes; the five recorded exceptions are not excluded from their final dynamic bounds. The profit targets are absolute units of the displayed per-firm payoff. Raw

TABLE VI
EVERY MISSING PURE FITTED ONE-STEP EQUILIBRIUM IN R7

Market	Seed	Time index	K_1	K_2	Minimum defect
1	11	1	0.140625	0.865625	6.492e-06
1	11	1	0.684375	0.140625	4.018e-07
1	11	10	0.321875	0.140625	1.538e-05
2	11	16	0.231250	1.046875	1.534e-05
2	29	13	0.140625	0.956250	1.394e-06

Minimum defect is the smallest maximum unilateral one-step improvement over all joint finite actions. These nodes remain in full dynamic verification; they are not called exact stage equilibria.

network policy, shortlist policy, and corrected finite lookup policy are different deployment objects. The principal R7 guard fraction is zero, but the full joint-action oracle was nevertheless constructed during training. A mixed-strategy *profile* was not estimated; the preceding argument explains why mixed unilateral deviations against the fixed pure profile are already covered by the finite best-response bound.

16. PAIRED PAYOFFS, COSTS, AND ECONOMIC SCALING

Each row below compares the two frozen methods on the same stored Brownian paths, at the indicated dimension, seed, fitting domain, and Euler resolution. Intervals are pointwise Student intervals across 512 pathwise differences. They are not independent across step sizes, because the Brownian increments are nested; no simultaneous-coverage claim is made. The narrow and wide policies were sep-

arately fitted, so their comparison also changes the training domain and update budget.

TABLE VII: All method-paired policy-payoff comparisons

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
10	11	Narrow	40	-0.03043	-0.03234	-0.02852
10	11	Narrow	80	-0.03237	-0.03438	-0.03036
10	11	Narrow	160	-0.03338	-0.03544	-0.03133
10	11	Wide	40	0.00061	0.00058	0.00064
10	11	Wide	80	0.00055	0.00052	0.00057
10	11	Wide	160	0.00052	0.00049	0.00054
10	29	Narrow	40	-0.02800	-0.02946	-0.02655
10	29	Narrow	80	-0.02993	-0.03146	-0.02840
10	29	Narrow	160	-0.03094	-0.03250	-0.02938
10	29	Wide	40	-0.00007	-0.00008	-0.00006
10	29	Wide	80	-0.00004	-0.00005	-0.00003
10	29	Wide	160	-0.00003	-0.00004	-0.00002
10	47	Narrow	40	-0.02522	-0.02695	-0.02349
10	47	Narrow	80	-0.02675	-0.02856	-0.02493
10	47	Narrow	160	-0.02755	-0.02941	-0.02569
10	47	Wide	40	0.00004	0.00003	0.00005
10	47	Wide	80	0.00001	0.00001	0.00002
10	47	Wide	160	0.00000	-0.00001	0.00001
20	11	Narrow	40	-0.05798	-0.06054	-0.05543
20	11	Narrow	80	-0.06131	-0.06396	-0.05866
20	11	Narrow	160	-0.06302	-0.06572	-0.06033
20	11	Wide	40	0.00123	0.00118	0.00128
20	11	Wide	80	0.00097	0.00092	0.00101
20	11	Wide	160	0.00084	0.00080	0.00088
20	29	Narrow	40	-0.01308	-0.01346	-0.01269
20	29	Narrow	80	-0.01401	-0.01442	-0.01361
20	29	Narrow	160	-0.01449	-0.01490	-0.01408
20	29	Wide	40	0.00480	0.00463	0.00496

Dim.	Seed	Domain	Steps	NBO – direct	Lower	Upper
20	29	Wide	80	0.00432	0.00417	0.00448
20	29	Wide	160	0.00409	0.00394	0.00424
20	47	Narrow	40	-0.19715	-0.19765	-0.19666
20	47	Narrow	80	-0.20011	-0.20061	-0.19962
20	47	Narrow	160	-0.20161	-0.20211	-0.20111
20	47	Wide	40	0.00084	0.00078	0.00089
20	47	Wide	80	0.00066	0.00061	0.00071
20	47	Wide	160	0.00058	0.00053	0.00062

Notes. Pointwise 95 percent Monte Carlo intervals, 512 paired paths per row. They do not bound optimality, continuous-time bias, or simultaneous coverage. The additional representation comparisons below use the identical NDU finite model. Gated regression has more parameters and higher per-update work than the width-32 tanh critic. Tensor projection exploits low state dimension. Both facts are explicit rather than hidden by matching only nominal iteration counts. The R7 NDU actor’s

TABLE VIII

ADDITIONAL CONTINUATION REPRESENTATIONS, IDENTICAL FINITE PREFERENCE ECONOMY

Method	Seed	Parameters	Policy bound	Value error	Train (s)	Verify (s)
Chebyshev	11	121	0.0121	0.0775	0.14	4.41
Chebyshev	11	225	0.0039	0.0165	0.15	4.48
Chebyshev	11	49	0.0373	0.2325	0.14	4.36
Gated	11	9345	0.0079	0.0284	41.28	4.12
Gated	29	9345	0.0098	0.0200	42.59	4.49
Gated	47	9345	0.0088	0.0306	38.99	4.30

Gated denotes a DGM-style architecture fitted to finite Bellman targets, not a differential DGM solver. Chebyshev degrees are 6, 10, and 14. All actions are enumerated; construction and reference costs are separately retained in JSON. Deterministic projections are not replicated under fictitious random seeds.

correction map stores a small number of changed action indices, but its complete

policy table occupies 68,000 bytes with the archived int64 representation. A minimal pair of int64 state-time and replacement indices requires 16 bytes per correction, excluding indexing and container overhead. A stored-table deployment avoids actor evaluation at existing nodes; a raw-network deployment requires its network weights and argmax calculation. Neither observation establishes off-grid policy or guard correctness. The continuous hybrid additionally requires local action queries using the stored continuation representation.

In the continuous-action study, training-query counts exclude verification queries by construction. All-action references are created only after policies and critics have been frozen. The raw actor, direct-gradient, multi-head, and smoke failures are retained with their full configurations. The complete action verifier’s tighter and looser implementations have separate records, including unresolved boxes and declared budgets. A passing regression suite means that arithmetic, scope, and accounting checks passed; it does not change a failed economic accuracy target into a success.

TABLE IX

RETAINED CONTINUOUS-ACTION DEVELOPMENT FAILURES

Configuration	Seed	Deviation gain	Train (s)	Target .1
Actor smoke	11	0.1872	0.53	Fail
Raw actor	11	0.2596	15.77	Fail
Four-head reset	11	1.1033	16.83	Fail
Direct gradient	11	0.1997	16.07	Fail

All executed pilot sources, raw arrays, and histories remain. The smoke grid and budget differ from the principal grid; it is not a principal-seed replicate. Pilot selection and subsequent improvements are explicitly post-review development.

17. HISTORICAL NUMERICAL RECORD AND SOURCE PRESERVATION

The preceding manuscript and supplement are archived byte-for-byte under the R8 revision directory. Every earlier economic equation, application, numerical source, raw array, failed run, and referee report remains in the repository history and the candidate tree. The main text now presents the successful R7 finite-policy

computations and the new continuous-action study at the point where the corresponding economic model is introduced. Earlier R6 policy failures and non-neural game results are not relabelled as current neural results. Their original discussions are reproduced below as dated historical material.

17.1. *R6 preference-policy discussion (historical)*

17.2. *Neural solution and action-domain sensitivity*

We now apply multilayer NBO to the same two-state economy, with the original cardinal utility, correlation, reflection rule, and stopping payoff unchanged. At each backward time level a two-hidden-layer, width-thirty-two tanh critic approximates its own Bellman continuation; a multilayer categorical actor selects among feasible actions. The reference solution is computed separately and is not used as a training target. The actor is fitted before policy evaluation at that time level, with the next-period critic fixed. The direct comparator eliminates the actor by enumeration of the same action set. Both methods have a cap of 160 L-BFGS inner iterations per time level, divided 100/60 between critic and actor for NBO.

An initial unweighted classification implementation produced a large policy loss. Absorbing wealth-face labels, for which every action has identical value, contaminated the policy fit; classification error also ignored the economic cost of a mistaken low-consumption action. The retained implementation excludes those faces from actor training and minimizes expected normalized Bellman action loss, with a small cross-entropy term. The pilot source, weights, and failed output are preserved. This change repairs an implementation failure rather than redefining the economic objective: the deployed action is still evaluated against *all* finite feasible alternatives.

For the principal comparison, $(n_u, n_x, N_t) = (17, 25, 20)$ and $k = 1$. The expanded action set contains 175 triples, with $c/X \in \{.02, .05, .10, .20, .30\}$, $p \in \{0, .375, .75, 1.125, 1.5\}$, and $\theta \in \{-.45, -.30, -.15, 0, .15, .30, .45\}$. Every time–state–action combination enters the error accounts in the main-text Bellman error accounts. Independently solving the fixed learned policy gives its actual regret relative to the finite reference. Midpoint value errors and derivative comparisons are additional

TABLE X

NEURAL PREFERENCE ADJUSTMENT ON THE SAME FINITE ECONOMY

Method	Seed	Value error	Policy regret	Bound R_0	Midpoint RMS
NBO	11	0.1679	0.1656	3.3954	0.0438
NBO	29	0.1495	0.1487	2.8503	0.0413
NBO	47	0.1572	0.1014	3.0605	0.0445
Direct Bellman	11	0.1162	0.0261	1.7944	0.0477
Direct Bellman	29	0.1061	0.0261	1.6930	0.0400
Direct Bellman	47	0.1342	0.0269	1.8224	0.0424

The first three error columns use every state and time of the $17 \times 25 \times 21$ grid with 175 actions; R_0 is the time-zero certificate, while the first two columns maximize their independently computed errors over all times. Midpoints compare the time-zero neural critic with the finite-reference interpolant. Continuous-action and controlled-diffusion discretization errors are not included. The accuracy targets were value error .2 and policy regret .1.

diagnostics, not continuous-domain certificates. The three-seed record contains a failed NBO accuracy target and reports it without exclusion. The direct comparator is more accurate at this budget. A categorical actor therefore has a measurable approximation cost even when the critic class is identical. Neural representation alone is not a reason to replace an inexpensive two-state grid solver; the grid here supplies an unusually complete audit of the neural computation.

Action bounds are part of the economy, not innocuous numerical constants. The sensitivity study reproduces the referee's action-set comparison and extends it to 1,053 actions in the same expanded box, a wider consumption–portfolio–adjustment box, a wider wealth domain, and two finer space–time grids, for both $k = 1$ and $k = 5$. The table in the supplement records lower- and upper-bound frequencies for every control. Consumption and portfolio bounds remain frequently active after the first expansion. Thus state-grid convergence of the old 27-action model does not establish convergence to an unconstrained control optimum. The wider sets describe different constrained economies; refinement within a fixed box is a different comparison. No continuous-action coverage or controlled-diffusion discretization

TABLE XI

DYNAMIC COURNOT: INDEPENDENT FIXED-RIVAL BEST RESPONSES

Grid	Steps	Actions	M_2	$I_1(0)$	$V_1(0)$	Max. gain
17^2	30	11	1	0	1.3852	0
17^2	30	11	2	0.4800	2.6737	0
25^2	60	17	1	0	1.3905	0
25^2	60	17	2	0.5500	2.6610	0

Center state is $(K_1, K_2) = (.775, .775)$. Horizon is three, capital lies in $[.05, 1.5]^2$, and investment in $[0, .8]$.

The final column maximizes independently computed dynamic unilateral gains over both firms and every finite-model state and time. Zero denotes the displayed floating-point precision; it is not a continuous-game error bound. Six principal cases additionally include a high-index equilibrium tie rule and asymmetric adjustment costs.

error is set to zero in the finite certificate. This preserves the economic application while making its constraint dependence observable.

17.3. R6 dynamic-game discussion (historical)

17.4. A dynamic capital-game computation

For two firms, we compute the finite-horizon capital game above with $A = 1$, $P_0 = 2$, $\delta = .1$, $\sigma_K = .15$, $\rho = .04$, $T = 3$, investment in $[0, .8]$, and capital in $[.05, 1.5]^2$. Terminal or joint-domain exit pays $.5K_i$. The baseline quadratic adjustment coefficients are $b_1 = b_2 = 1$; an asymmetric experiment uses $b_2 = 1.2$. Independent diffusion cubature and positive interpolation define the finite dynamic game. Investment nets include both bounds.

Backward induction constructs the two continuation-payoff matrices at every state and time. The solver enumerates pure stage equilibria; alternative deterministic tie rules are retained. If no pure equilibrium exists, the solver records this fact and the smallest maximum stage deviation it finds, rather than declaring equilibrium. After fixing the entire computed rival policy, a *separate backward optimization* permits every own action at every state and time. Its value minus the profile payoff is the unilateral exploitability map. A one-period check with a frozen continuation value alone would not perform this test. All six principal finite-game cases have

pure equilibria at every stage and zero detected exploitability to the reported arithmetic precision. A deliberately coarse regression case has a missing pure stage equilibrium and is reported as approximate, with time-indexed deviation bounds. The table includes fixed-market $M_2 = 1$ and growing-market $M_2 = 2$ normalizations as distinct economies; the resulting investment differences do not prove an $N \rightarrow \infty$ limit. Binding-investment frequencies, alternative equilibrium selections, and the asymmetric case are in the raw record. This is an actual dynamic best-response computation, not the static $1/64$ diagnostic. It is a finite-grid game result; a trained neural game and a continuous-state equilibrium bound are different objects and are not inferred from it.

18. COMPLETE R9 REFINEMENT AND VERIFICATION ACCOUNTS

This section retains every configuration used to assess the new signed cover and guarded neural map. The preceding R6–R8 sections are historical evidence, not overwritten runs. The new common and fine-grid envelopes must not be interchanged: they concern different state-time discretizations. Continuous-state/time defects remain separately unspecified for the stopped preference diffusion.

TABLE XII

FROZEN-POLICY ERRORS OVER EVERY REFINEMENT NODE

Policy	Grid/dates	Max.	95th	99th	t	u	X
A11	17/25/20	0.0612	0.0023	0.0073	0.000	1.99	0.530
A11	25/37/40	0.5360	0.2255	0.3885	0.000	3.00	0.540
A11	33/49/80	0.6550	0.2647	0.5264	0.000	3.00	0.530
A29	17/25/20	0.0143	0.0013	0.0038	0.550	1.20	1.782
A29	25/37/40	0.4959	0.2178	0.4154	0.000	1.27	1.587
A29	33/49/80	0.7011	0.2552	0.5493	0.000	1.26	1.587
A47	17/25/20	0.0467	0.0048	0.0073	0.000	1.54	0.530
A47	25/37/40	0.5371	0.2040	0.3549	0.000	3.00	0.540
A47	33/49/80	0.6562	0.2484	0.4698	0.000	3.00	0.530
D11	17/25/20	0.0230	0.0026	0.0047	0.400	1.20	1.782
D11	25/37/40	0.9249	0.3603	0.5999	0.000	2.85	0.540
D11	33/49/80	1.2997	0.4389	0.7763	0.000	2.89	0.530
D29	17/25/20	0.0157	0.0013	0.0025	0.550	1.20	1.782
D29	25/37/40	1.5662	0.3699	0.5710	0.000	3.00	0.520
D29	33/49/80	2.1234	0.4423	0.7166	0.163	3.00	0.515
D47	17/25/20	0.0231	0.0034	0.0057	0.800	1.20	0.595
D47	25/37/40	0.7631	0.3158	0.4675	0.250	2.92	0.540
D47	33/49/80	0.9957	0.3882	0.6275	0.000	2.94	0.530

A denotes actor-plus-search and D direct-plus-search. Quantiles concern the positive reference advantage over every state and date. Maximum locations include stopping-adjacent states. All eighteen historical configurations are retained; no centre-only selection is used.

TABLE XIII

BOUNDARY-REGION REFINEMENT DIAGNOSTICS

Policy	Grid/dates	Lower region	Upper region	Interior	Above 0.1, %
A11	17/25/20	0.0612	0.0211	0.0176	0.00
A11	25/37/40	0.5360	0.4306	0.4604	11.61
A11	33/49/80	0.3635	0.5851	0.6550	14.27
A29	17/25/20	0.0106	0.0143	0.0092	0.00
A29	25/37/40	0.4473	0.4204	0.4959	10.22
A29	33/49/80	0.2698	0.5630	0.7011	13.21
A47	17/25/20	0.0467	0.0181	0.0141	0.00
A47	25/37/40	0.5371	0.4080	0.4458	11.07
A47	33/49/80	0.3641	0.4891	0.6562	13.13
D11	17/25/20	0.0147	0.0230	0.0114	0.00
D11	25/37/40	0.9249	0.4434	0.7900	18.70
D11	33/49/80	1.1417	0.5408	1.2997	21.20
D29	17/25/20	0.0146	0.0157	0.0090	0.00
D29	25/37/40	1.5662	0.4878	1.1079	18.60
D29	33/49/80	2.1234	0.5859	1.9784	20.80
D47	17/25/20	0.0149	0.0192	0.0231	0.00
D47	25/37/40	0.7631	0.4937	0.6036	16.52
D47	33/49/80	0.9140	0.5868	0.9957	18.73

Each wealth boundary region spans two grid cells. Interior means the remaining wealth nodes. Values are maximum finite-reference advantages, not upper certificates.

TABLE XIV

REFINEMENT ACTION SATURATION RATES, PERCENT

Policy	Grid/dates	c_-	p_-	θ_-	c_+	p_+	θ_+
A11	17/25/20	0.0	0.4	28.4	96.8	53.3	41.0
A11	25/37/40	0.0	0.1	23.4	95.7	47.7	36.0
A11	33/49/80	0.0	0.2	24.2	95.7	48.0	37.6
A29	17/25/20	0.0	0.1	28.1	96.9	54.3	40.7
A29	25/37/40	0.0	0.0	22.6	95.8	49.0	35.5
A29	33/49/80	0.0	0.0	23.4	95.8	49.0	37.2
A47	17/25/20	0.0	0.3	28.1	97.0	54.5	41.1
A47	25/37/40	0.0	0.1	23.2	95.7	49.1	36.3
A47	33/49/80	0.0	0.1	24.0	95.9	49.2	37.7
D11	17/25/20	0.0	0.1	27.3	97.4	53.3	41.8
D11	25/37/40	0.0	0.0	21.6	91.4	46.9	34.5
D11	33/49/80	0.0	0.0	22.8	92.8	47.5	36.4
D29	17/25/20	0.0	0.0	27.5	97.4	54.5	41.5
D29	25/37/40	0.0	0.0	21.7	91.4	48.6	34.2
D29	33/49/80	0.0	0.0	22.9	92.8	48.7	36.2
D47	17/25/20	0.0	0.1	27.5	97.6	54.7	41.8
D47	25/37/40	0.0	0.0	21.8	91.6	48.7	34.5
D47	33/49/80	0.0	0.0	23.0	93.1	48.9	36.5

Rates cover active state-time nodes; an action is counted within 10^{-4} of a bound. The box is an economic primitive, and saturation is not interpreted as an unconstrained first-order optimum.

19. FRESH TRAINING, CORRECTION STORAGE, AND TOTAL COST

The new spatial refinement is genuinely nested: every coarse spatial node is a fine spatial node, and every coarse date is a fine date. A trained fine-grid policy is not substituted into the frozen-policy historical experiment. The guard acts only on the chosen policy’s evaluation target. Its storage and query cost are explicit, and the actor-free comparison receives the same correction.

TABLE XV

EVERY FRESH NESTED-GRID TRAINING RUN

Policy	Nodes u /dates	Ref. max.	Value fit	Seconds	Corrected %
A11G	17/20	0.0372	0.0002	26.35	90.5
A29G	17/20	0.0085	0.0003	27.12	90.1
A47G	17/20	0.0372	0.0003	25.72	90.4
S11G	17/20	0.0372	0.0003	13.02	90.6
S29G	17/20	0.0372	0.0003	13.26	89.9
S47G	17/20	0.0371	0.0003	12.81	90.0
A11	33/40	0.0932	0.2043	89.58	–
A29	33/40	0.0625	0.1650	89.60	–
A47	33/40	0.0872	0.1649	85.80	–
S11	33/40	0.0932	0.1605	56.66	–
S29	33/40	0.0626	0.1910	50.84	–
S47	33/40	0.0932	0.1545	45.71	–
A11G	33/40	0.0041	0.0001	88.85	95.0
A29G	33/40	0.0093	0.0001	85.55	95.1
A47G	33/40	0.0155	0.0001	88.86	95.0
S11G	33/40	0.0138	0.0001	47.69	95.0
S29G	33/40	0.0156	0.0001	49.15	95.1
S47G	33/40	0.0156	0.0001	45.36	95.2

G denotes the indexed guard. All eighteen fresh runs appear. Value fit is a policy-value diagnostic, distinct from the one-step target residual. Unguarded failures are not excluded.

TABLE XVI

INDEPENDENT VERIFICATION AND CORRECTION STORAGE

Policy	Uniform bound	Policy residual	Correction KiB	Evaluation s.
A11	0.0963	0.1100527	0.0	4.22
A29	0.0654	0.0962556	0.0	4.20
A47	0.0903	0.1126844	0.0	4.48
S11	0.0963	0.1103166	0.0	5.01
S29	0.0655	0.0967997	0.0	4.99
S47	0.0963	0.1130800	0.0	5.00
A11G	0.0140	0.0000623	568.5	4.20
A29G	0.0311	0.0000625	568.5	4.24
A47G	0.0356	0.0000624	568.5	4.28
S11G	0.0354	0.0000624	568.5	5.01
S29G	0.0357	0.0000622	568.5	5.01
S47G	0.0349	0.0000624	568.5	5.01

Same fine-grid upper envelope for every row. Correction memory counts saved indexed corrections and masks over all dates, not only one network. Policy residuals use independent outward point-action evaluation. The complete source contains raw arrays and nodewise masks.

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32

1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17
18
19
20
21
22
23
24
25
26
27
28
29
30
31
32

TABLE XVII
EXPLICIT SHARED-COST ACCOUNTING, SECONDS

Policy	Train	One policy	Nine policies	100 policies	Nine + welfare
A11	15.01	389.04	57.91	20.24	85.32
A29	28.49	402.57	71.44	33.77	98.30
A47	26.46	400.53	69.40	31.73	96.34
D11	32.02	406.18	75.05	37.38	102.10
D29	32.80	406.98	75.84	38.17	104.18
D47	30.52	404.74	73.60	35.93	101.10
S11	13.97	388.27	57.14	19.47	88.45
S29	13.72	388.06	56.92	19.25	88.26
S47	12.87	387.20	56.06	18.39	87.43

Training/reference/policy-evaluation times are archived R8 measurements; the new certificate and welfare measurements are current. This is an auditable mixed-provenance cost account, not a same-machine speed comparison. Each policy receives its own independent payoff interval. Shared-cover cost is amortized only within the identical economy.

20. AN ABSOLUTE BENCHMARK AND PAIRED CAPITAL PAYOFFS

The continuous analytical bound in the article applies to the original unbounded-state capital economy from the specified initial state. The simulations below assess an explicit feasible schedule on the same discretized Brownian paths as the historical learned policies. A narrow Monte Carlo interval cannot remove unknown Euler bias. The analytical bound and the simulation comparison have different, stated roles.

TABLE XVIII

FEASIBLE SCHEDULE ON THE MATCHED BROWNIAN PATHS

Dimension	Euler steps	Mean payoff	95% lower	95% upper
10	40	-1.2555	-1.2694	-1.2416
10	80	-1.2583	-1.2723	-1.2444
10	160	-1.2598	-1.2738	-1.2458
20	40	-1.2565	-1.2697	-1.2433
20	80	-1.2595	-1.2728	-1.2463
20	160	-1.2611	-1.2744	-1.2478

512 paths per dimension. Pointwise Student intervals describe Monte Carlo error only; no continuous-time bias correction is assumed.

TABLE XIX

LEARNED-POLICY MINUS FEASIBLE-SCHEDULE PAYOFFS

Dimension	Policy	Training domain	Mean	95% lower	95% upper
10	N11	Narrow	-0.0544	-0.0572	-0.0517
10	N11	Wide	0.0008	0.0008	0.0008
10	N29	Narrow	-0.0490	-0.0510	-0.0470
10	N29	Wide	0.0008	0.0008	0.0009
10	N47	Narrow	-0.0452	-0.0476	-0.0429
10	N47	Wide	0.0003	0.0002	0.0003
10	D11	Narrow	-0.0211	-0.0218	-0.0203
10	D11	Wide	0.0003	0.0002	0.0003
10	D29	Narrow	-0.0181	-0.0185	-0.0176
10	D29	Wide	0.0009	0.0008	0.0009
10	D47	Narrow	-0.0177	-0.0182	-0.0171
10	D47	Wide	0.0003	0.0002	0.0003
20	N11	Narrow	-0.0822	-0.0854	-0.0791
20	N11	Wide	0.0003	0.0002	0.0003
20	N29	Narrow	-0.0256	-0.0263	-0.0249
20	N29	Wide	0.0003	0.0003	0.0003
20	N47	Narrow	-0.2429	-0.2441	-0.2417
20	N47	Wide	0.0004	0.0004	0.0004
20	D11	Narrow	-0.0192	-0.0197	-0.0187
20	D11	Wide	-0.0006	-0.0006	-0.0005
20	D29	Narrow	-0.0111	-0.0114	-0.0108
20	D29	Wide	-0.0038	-0.0039	-0.0036
20	D47	Narrow	-0.0413	-0.0423	-0.0403
20	D47	Wide	-0.0002	-0.0002	-0.0001

Every stored configuration is retained. Differences are paired on the same 160-step Euler paths; confidence intervals do not cover Euler discretization bias. They are not continuous neural-policy regret bounds.